MorphGrower: A Synchronized Layer-by-layer Growing Approach for Plausible Neuronal Morphology Generation

Proceedings of the 41st International Conference on Machine Learning (ICML 2024), PMLR 235, Vienna, Austria

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Code: https://github.com/Thinklab-SJTU/MorphGrower

Method Overview: What We Propose

- Layer-by-layer generation; sibling branches decoded in synchronized pairs.
- Each layer conditioned on prior structure: local path + global previous layers.
- CVAE with vMF latent; branch LSTMs; global forest GNN+GRU; local EMA.
- Soma layer: provided or via unconditional VAE (†).

Architecture diagram (Fig. 2): CVAE with branch LSTMs, global forest GNN+GRU, local EMA, paired decoder

Architecture (Fig. 2): branch LSTMs + global forest encoder (GNN+GRU) + local EMA feed a CVAE (vMF latent); decoder outputs sibling branches jointly.

Generation snapshot: reference layer, generated subtree, newly synthesized layer

Layer-wise inference: condition on generated subtree and reference layer to synthesize the next layer.

Qualitative Examples: Realistic, Valid Morphologies

Side-by-side xy/xz/yz projections of real vs generated neurons across datasets

Consecutive snapshot strip: blue previous subtree, pink new layer for RGC

Sibling-branch pairing visualization with annotations at a bifurcation

Qualitative Montages Across Neuron Types

VPM montage row 1

VPM montage row 2

RGC montage row 1

RGC montage row 2

M1-EXC montage row 1

M1-EXC montage row 2

M1-INH montage row 1

M1-INH montage row 2

Zoomed Qualitative Details: Tips and Bifurcations

Zoomed patches highlighting branch tips and bifurcations with overlays

Takeaway & Contact

- Synchronized, layer-wise paired-branch growth yields valid, realistic morphologies conditioned on prior structure.

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